

BedForge QC — Blueprint Intelligence

Blueprint Assessment Pack

Internal verification artifact. Compare side-by-side with plant shop drawings. Prestressed plant branding is secondary.

A. Cover / identity

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Controlled fail reasons

- Critical fields require manual verification before lock: overall_depth_in, overall_length_ft.

B. Critical identity

Large first-pass identity for plant-job comparison (Job L25390 / CID / Type 2 / marks / lengths). Schema-absent keys are listed explicitly so missing DNA is visible.

Job number (job_number)	L25390	medium	confirmed	1	Job number not confidently located.
Beam mark(s) (beam_mark)	s	medium	confirmed	1	Beam mark not confidently located.
Product family / beam type (product_family)	i_beam	high	confirmed	1	—
County / DOT (county_dot)	, Kentucky	medium	confirmed	1	County/DOT owner not confidently located.
bridge_id	— (not in current extractor schema)	—	not_in_schema	—	Bridge ID / CID is not extracted by controlled_regex_v1.
route	— (not in current extractor schema)	—	not_in_schema	—	Route is not extracted by controlled_regex_v1.
Overall length (overall_length_ft)	—	low	unconfirmed	—	Overall length not confidently located.

casting_length_ft	— (not in current extractor schema)	—	not_in_schema	—	Casting length is not extracted by controlled_regex_v1.
Depth (overall_depth_in)	—	low	unconfirmed	—	Overall depth/height not confidently located.

C. Full field inventory

Every FIELD_GROUPS key currently known to Blueprint Intelligence, including empty and not-applicable fields.

C1. Title Block / Identity (title_block)

job_number	L25390	medium	confirmed	1	Job number not confidently located.
project_name	—	low	unconfirmed	—	Project name not confidently located.
county_dot	, Kentucky	medium	confirmed	1	County/DOT owner not confidently located.
sheet_number	—	low	unconfirmed	—	Sheet number not confidently located.
revision	—	low	unconfirmed	—	Revision not confidently located.
beam_mark	s	medium	confirmed	1	Beam mark not confidently located.
product_family	i_beam	high	confirmed	1	—
overall_length_ft	—	low	unconfirmed	—	Overall length not confidently located.
design_camber_in	—	low	unconfirmed	—	Design camber note not clearly stated.
finish_notes	—	low	unconfirmed	—	Finish notes present in drawing text; verify manually.

C2. Geometry (geometry)

overall_depth_in	—	low	unconfirmed	—	Overall depth/height not confidently located.
top_flange_width_in	—	low	unconfirmed	—	Top flange width not confidently located.
top_flange_thickness_in	—	low	unconfirmed	—	Top flange thickness not confidently located.
bottom_flange_width_in	—	low	unconfirmed	—	Bottom flange width not confidently located.
bottom_flange_thickness_in	—	low	unconfirmed	—	Bottom flange thickness not confidently located.
web_thickness_in	—	low	unconfirmed	—	Web thickness not confidently located.
outer_width_in	—	low	not_applicable	—	Not required for I-beam translation unless the drawing explicitly calls it out.

outer_depth_in	—	low	not_applicable	—	Not required for I-beam translation unless the drawing explicitly calls it out.
wall_thickness_in	—	low	not_applicable	—	Not required for I-beam translation unless the drawing explicitly calls it out.
void_width_in	—	low	not_applicable	—	Not required for I-beam translation unless the drawing explicitly calls it out.
void_depth_in	—	low	not_applicable	—	Not required for I-beam translation unless the drawing explicitly calls it out.
special_end_geometry	—	low	unconfirmed	—	Special end geometry/end-detail notes present; verify manually.

C3. Strand System (strand_system)

strand_count	—	low	unconfirmed	—	Strand count not confidently located.
strand_diameter_in	—	low	unconfirmed	—	Strand diameter not confidently located.
strand_area_in2	—	low	unconfirmed	—	Strand area not confidently located.
strand_pattern_rows	—	low	unconfirmed	—	—
straight_strand_count	—	low	unconfirmed	—	Straight strand count not clearly stated.
draped_strand_count	—	low	unconfirmed	—	Draped strand count not clearly stated.
hold_downs	—	low	unconfirmed	—	No deterministic hold-down station callouts were found in extractable text.
jacking_force_kip	—	low	unconfirmed	—	Jacking force note not confidently located.
target_elongation_in	—	low	unconfirmed	—	Target elongation note not confidently located.
debond_notes	—	low	unconfirmed	—	Debonded or shielded strand notes present; verify manually.

C4. Hardware (hardware)

lift_loops	—	low	unconfirmed	—	No deterministic lift loop station callouts were found in extractable text.
inserts	—	low	unconfirmed	—	No deterministic insert station callouts were found in extractable text.
tubes	—	low	unconfirmed	—	No deterministic tube station callouts were found in extractable text.

tie_rod_openings	—	low	unconfirmed	—	No deterministic tie-rod opening station callouts were found in extractable text.
drain_holes	—	low	unconfirmed	—	No deterministic drain hole station callouts were found in extractable text.
stirrups	—	low	unconfirmed	—	Only stirrup spacing was auto-detected. Region/type still require review.
plates_and_clips	—	low	unconfirmed	—	Plate, clip, or bearing notes present; verify manually.
grout_grooves	—	low	not_applicable	—	Not required for I-beam translation unless the drawing explicitly calls it out.
bituminous_ends	—	low	unconfirmed	—	Bituminous or cut-off pocket treatment noted; verify manually.

C5. Ends, QC & Finishes (ends_and_qc)

marked_end_rule	—	low	unconfirmed	—	Marked end rule not confidently located.
end_elevation_notes	—	low	unconfirmed	—	End elevation note present; verify manually.
paint_id_requirements	Paint or ID marking note present; verify manually.	medium	confirmed	1	Captured from keyword hit; verify exact wording against source.
dimensional_tolerances	—	low	unconfirmed	—	Tolerance note present; verify manually.
special_inspection_notes	Special inspection note present; verify manually.	medium	confirmed	1	Captured from keyword hit; verify exact wording against source.

D. Missing / weak fields

CRITICAL_FIELDS that are empty, low-confidence, or UNCONFIRMED. These block or weaken lock / twin use.

- **overall_depth_in — value=— | conf=low | status=unconfirmed | page=— | Overall depth/height not confidently located.**
- **overall_length_ft — value=— | conf=low | status=unconfirmed | page=— | Overall length not confidently located.**

Other empty / unconfirmed / low-confidence fields

- bituminous_ends [unconfirmed/low] page=— value=—
- bottom_flange_thickness_in [unconfirmed/low] page=— value=—
- bottom_flange_width_in [unconfirmed/low] page=— value=—
- debond_notes [unconfirmed/low] page=— value=—
- design_camber_in [unconfirmed/low] page=— value=—
- dimensional_tolerances [unconfirmed/low] page=— value=—
- drain_holes [unconfirmed/low] page=— value=—
- draped_strand_count [unconfirmed/low] page=— value=—
- end_elevation_notes [unconfirmed/low] page=— value=—
- finish_notes [unconfirmed/low] page=— value=—
- grout_grooves [not_applicable/low] page=— value=—
- hold_downs [unconfirmed/low] page=— value=—
- inserts [unconfirmed/low] page=— value=—
- jacking_force_kip [unconfirmed/low] page=— value=—
- lift_loops [unconfirmed/low] page=— value=—
- marked_end_rule [unconfirmed/low] page=— value=—
- outer_depth_in [not_applicable/low] page=— value=—
- outer_width_in [not_applicable/low] page=— value=—
- plates_and_clips [unconfirmed/low] page=— value=—
- project_name [unconfirmed/low] page=— value=—
- revision [unconfirmed/low] page=— value=—
- sheet_number [unconfirmed/low] page=— value=—
- special_end_geometry [unconfirmed/low] page=— value=—
- stirrups [unconfirmed/low] page=— value=—
- straight_strand_count [unconfirmed/low] page=— value=—
- strand_area_in2 [unconfirmed/low] page=— value=—
- strand_count [unconfirmed/low] page=— value=—
- strand_diameter_in [unconfirmed/low] page=— value=—
- strand_pattern_rows [unconfirmed/low] page=— value=—
- target_elongation_in [unconfirmed/low] page=— value=—
- tie_rod_openings [unconfirmed/low] page=— value=—
- top_flange_thickness_in [unconfirmed/low] page=— value=—
- top_flange_width_in [unconfirmed/low] page=— value=—
- tubes [unconfirmed/low] page=— value=—
- void_depth_in [not_applicable/low] page=— value=—
- void_width_in [not_applicable/low] page=— value=—
- wall_thickness_in [not_applicable/low] page=— value=—
- web_thickness_in [unconfirmed/low] page=— value=—

Schema gaps vs L25390-class shop drawings (not extracted today)

- bridge_id / CID (e.g. 25-5390)
- route
- casting_length_ft (e.g. 47'-3" vs 52'-0")
- multi-mark schedule (201–209)
- standard section name (Type 2 I-Beam)
- strand grade (270 ksi)
- bed layout / bulkheads
- OCR for image-only sheets

E. Page text evidence appendix

OCR is not implemented. controlled_regex_v1 reads the PDF text layer only (pypdf extract_text).

Page 1

Text-layer excerpt (420 chars):

PRESTRESS SERVICES INDUSTRIES LLC
Flattened Shop Drawings – BedForge Upload Pack
Job No: L25390 | CID: 25-5390
KY 210 over North Fork Nolin River – Larue County, Kentucky
Bridge ID: 062B00034R | Type 2 I-Beam | Contractor: JAVE, LLC
Beam Marks: 201-209 (9 beams)
Images enhanced: contrast, sharpness, crop, orientation for AI extraction
BedForge QC development corpus · Flattened photo captures of accepted shop drawings

Page 2

Text-layer excerpt (182 chars):

L25390 · Sheet 1 of 17 · Cover Sheet / General Notes / Drawing Index
Job L25390 · Type 2 I-Beam · Larue County KY
2/10
Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

Page 3

Text-layer excerpt (158 chars):

L25390 · Sheet 2 of 17 · Erection Layout & Details
Beams 201-209 · Spans 1-3 · Diaphragms
3/10
Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

Page 4

Text-layer excerpt (196 chars):

L25390 · Sheet 3 of 17 · Casting Bed Layout (Span 2) + Strand Pattern Pours 1 & 2
MK 204/205/206 · 270' Bed · 1/2" 270K strands
4/10
Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

Page 5

Text-layer excerpt (200 chars):

L25390 · Sheet 4 of 17 · Casting Bed Layout (Spans 1 & 3) + Strand Pattern Pours 3 & 4
MK 201-209 bed positions · Hold-downs H-56-S
5/10
Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

Page 6

Text-layer excerpt (170 chars):

L25390 · Sheet 5 of 17 · Reinforcement Details / Bar Bending
#401(E) #403 #301 #302 · ASTM A615 Gr 60
6/10
Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

Page 7

Text-layer excerpt (162 chars):

L25390 · Sheet 6 of 17 · Beam Shop Drawing – Marks 201 / 202 / 203
47'-3" overall · Qty 1/3/1
7/10
Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

Page 8

Text-layer excerpt (162 chars):

L25390 · Sheet 7 of 17 · Beam Shop Drawing – Marks 204 / 205 / 206
52'-0" overall · Qty 1/3/1
8/10
Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

Page 9

Text-layer excerpt (162 chars):

L25390 · Sheet 8 of 17 · Beam Shop Drawing — Marks 207 / 208 / 209

47'-3" overall · Qty 1/3/1

9/10

Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

Page 10

Text-layer excerpt (160 chars):

L25390 · Sheet 9 of 17 · Hardware — Mark 201

Lift loops · F-64 inserts · H-56-S hold-downs

10/10

Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

F. Twin DNA readiness

Fields that would currently drive a 3D twin after lock via `normalize_locked_blueprint()`. Draft extractions do not render a production twin until Verify & Lock.

Twin lock state	NOT LOCKED — twin stays draft/legacy until lock
Normalized length	0
Cross-section keys	["bottom_flange_thickness_in", "bottom_flange_width_in", "overall_depth_in", "top_flange_thickness_in", "top_flange_width_in", "web_thickness_in"]
Strand pattern rows	0
Lift loops / hold-downs	0 / 0
Locked revision id	—

Would drive twin today (extracted + confirmed/manually_confirmed)

- `product_family` — Selects I-beam vs box-beam cross-section family

Required twin inputs still missing or unconfirmed

- `overall_length_ft` — Twin length (`normalized_blueprint.length`)
- `overall_depth_in` — I-beam overall depth
- `top_flange_width_in` — I-beam top flange width
- `top_flange_thickness_in` — I-beam top flange thickness
- `bottom_flange_width_in` — I-beam bottom flange width
- `bottom_flange_thickness_in` — I-beam bottom flange thickness
- `web_thickness_in` — I-beam web thickness
- `strand_pattern_rows` — Strand row DNA (else invented from counts)
- `strand_count` — Fallback strand count
- `straight_strand_count` — Fallback straight-row count
- `draped_strand_count` — Drape profile + draped row
- `hold_downs` — Hold-down stations / drape low points
- `lift_loops` — Lift-loop stations
- `inserts` — Insert stations
- `tubes` — Tube stations
- `tie_rod_openings` — Tie-rod stations
- `drain_holes` — Drain stations
- `stirrups` — Stirrup spacing (regions not modeled)
- `marked_end_rule` — Marked-end label
- `bituminous_ends` — Bituminous / cutoff pockets
- `jacking_force_kip` — Tension reference jacking force
- `target_elongation_in` — Tension reference elongation
- `strand_diameter_in` — Tension reference diameter
- `strand_area_in2` — Tension reference area

Normalized DNA snapshot (what lock would persist)

```
{"cross_section": {"overall_depth_in": null, "top_flange_width_in": null, "top_flange_thickness_in": null, "bottom_flange_width_in": null, "bottom_flange_thickness_in": null, "web_thickness_in": null}, "length": 0, "marked_end": {"label": "MARKED END", "end": "start"}, "bituminous_ends": [], "lift_loops": [], "inserts": [], "tubes": [], "tie_rod_openings": [], "drain_holes": [], "hold_downs": [], "grout_grooves": [], "dimensions": {}}
```